Multimodal-Augmented Semantic Identifiers for Cold-Start Discovery in Generative Recommendation: Research Proposal

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***Abstract*—The architecture of commercial recommendation systems is undergoing a paradigm shift. The traditional 'Retrieve-and-Rank' pipeline involves scoring millions of item embeddings. It is being replaced by Generative Recommendation (GR). GR frameworks unify discovery as a sequence-to-sequence generation task and leverage the auto-regressive capabilities of Large Language Models (LLMs). The efficacy of GR however, is bottlenecked by its indexing strategy (how items are encoded into discrete, generative tokens). Current state-of-the-art systems use Semantic Identifiers (SIDs). SIDs are derived from Residual Quantized Variational Autoencoders (RQ-VAE) that operate on textual metadata. Therefore, in visual-heavy domains like short-form video and e-commerce, this unimodal reliance creates a 'Semantic-Visual Gap'. However, recent studies highlight severe limitations in current multimodal SID extensions, specifically 'modality sensitivity collapse' and the 'semantic-behavioral gap'. Which might result in similar identifiers for visually distinct items. To tackle these problems, we propose the Multimodal-Augmented Semantic ID (MASI) framework. MASI abandons traditional early-fusion techniques in favor of a modality-specific Late-Fusion Codebook strategy, coupled with a Behavior-Aware Contrastive alignment phase. We aim to improve zero-shot, cold-start discovery by encoding both aligned semantic categorization and visual features into a unified generative token space.**

***Index Terms*—Generative Recommendation, SASRec, Semantic Identifiers, Multimodal Fusion, Cold-Start Discovery, RQ-VAE, Generative Retrieval.**

I. Introduction

Organizations like Meta, Amazon, Apple, Netflix, Google, and OpenAI process billions of interactions daily. Traditionally, these platforms operated on a multi-stage discriminative pipeline:

1) A fast retrieval model selects a few hundred candidates.
2) A heavy ranking model scores them.

However, the reasoning and generative capabilities of Large Language Models (LLMs) have catalyzed a shift towards Generative Recommendation (GR) [Liang and Y. Zhang 2025].

In GR, recommendation is not a classification or scoring problem. It is a conditional text generation task. Given a user's chronological interaction history, the model auto-regressively "writes" the identifier of the next item [Yang, H. Lin, Z. Zhang, et al. 2025]. The critical factor in this paradigm is the indexing strategy. Traditional systems used atomic IDs (e.g., assigning the integer `ID_1051` to a specific movie). These IDs are semantically hollow. When a new item is introduced (the cold-start problem), its atomic ID has no learned embedding, causing the system to fail.

To solve this, researchers introduced Semantic Identifiers (SIDs). SIDs are hierarchical codewords derived from item content (e.g., mapping a movie to a token sequence like `[14, 102, 12]`) [Singh et al. 2024].

SIDs however, are currently generated almost exclusively from textual descriptions. We argue that this creates a fundamental flaw in domains where visual features dictate preference. This proposal introduces the MASI framework, which is designed to encode both text and images into a single, unified generative token space.

II. Literature Survey: The Mechanics of Modern Recommendation Systems

This section provides a breakdown of the current state-of-the-art to understand the necessity of multimodal SIDs.

A. The Attention mechanism: Self-Attentive Sequential Rec (SASRec)

The foundational pillar of personalization before GR, was the modeling of sequential user behavior. Early approaches used Markov Chains, but the breakthrough was the Self-Attentive Sequential Recommendation (SASRec) model [Kang and McAuley 2018].

SASRec deploys the Transformer architecture for recommendation. Instead of words in a sentence, the input is represented as a sequence of item embeddings $E = [e_1, e_2, \ldots, e_n]$. The item embeddings represent a user's chronological clicks. SASRec employs multi-head self-attention to dynamically weigh historical interactions, allowing the model to capture both long-term preferences and short-term interactions without the rigid constraints posed by a Markov window. The attention mechanism computes a weighted sum of past items:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{1}$$

where Q, K, V are linear projections of the item embedding sequence.

The Flaw: SASRec treats prediction as a huge multi-class classification problem over the entire item catalog $|\mathcal{I}|$. At the final layer, it computes a dot product between the

user's sequence embedding and the item embedding matrix $\mathbf{E} \in \mathbb{R}^{|\mathcal{I}| \times d}$. This creates a huge memory bottleneck (complexity $O(|\mathcal{I}|)$). Moreover, as the item IDs are atomic one-hot vectors, a new item without any interaction history cannot be recommended (causing the cold-start problem).

B. The LLM Shift: Recommendation as Language Processing

To address the scaling and flexibility issues of SASRec, researchers proposed treating recommendation as a Natural Language Processing (NLP) task. The landmark P5 framework [Geng et al. 2022] demonstrated that recommendation could be unified within a standard encoder-decoder LLM (like T5).

In P5, the input is modeled as a natural language prompt template: *"User A watched 'Breaking Bad', 'Better Call Saul'. What will they watch next?"* The model outputs a text string: *"Narcos"*.

The Flaw: While conceptually elegant, P5 suffered from severe hallucination. Standard LLMs use sub-word tokenization (e.g., Byte-Pair Encoding). If an item is named "iPhone 15 Plus", the LLM might generate "iPhone 15 Pro" if the probabilistic token for "Pro" was slightly higher. Generating raw text strings is too unconstrained for precise item retrieval.

C. Hallucination's Solution: Generative Retrieval & Semantic IDs

To fix hallucination P5 suffered from, the community moved to Generative Retrieval, popularized by the TIGER framework [Rajput et al. 2023]. The model generates discrete, abstract tokens called 'Semantic Identifiers' (SIDs), instead of generating English text.

Each SID is structured as a hierarchical tuple, e.g., $S_i = \langle c_1, c_2, c_3 \rangle$. The hierarchy is strictly enforced: c_1 might represent a broad cluster (Electronics), c_2 a sub-cluster (mobile phones), and c_3 the specific item (Apple iPhone 15). The auto-regressive model is then trained on these discrete tokens, thereby reducing the generation space from $O(|\mathcal{I}|)$ to $O(D \times |\mathbb{V}|)$, where D is the tuple length and $|\mathbb{V}|$ is the small vocabulary size of the codebook. If the model generates a prefix $\langle c_1, c_2 \rangle$, it means that it has successfully narrowed the search space to a specific semantic cluster, thereby solving the hallucination problem and allowing for zero-shot recommendation of new items that share that prefix.

D. The Mathematical Engine: RQ-VAE

The current standard for generating SIDs is the Residual Quantized Variational Autoencoder (RQ-VAE) [Lee et al. 2022]. Unlike standard clustering algorithms (like K-Means), RQ-VAE learns a structured hierarchy of embeddings.

Given a continuous text embedding of an item $z \in \mathbb{R}^d$ (extracted via BERT), the RQ-VAE uses D sequential codebooks $C = \{C_1, \ldots, C_D\}$, where each codebook contains K trainable vectors.

1) **Level 1:** The model finds the closest vector to z in C_1. Let its index be c_1.
2) **Compute Residual:** The model subtracts the codebook vector from z to find the "error" or residual: $r_1 = z - C_1(c_1)$.
3) **Level 2:** The model then quantizes the residual r_1 using the second codebook C_2, yielding index c_2.

This process repeats D times. The final SID is the sequence of indices $(c_1, c_2, \ldots, c_D)$. Mathematically, this forces a tree structure:

- c_1 captures the largest variance (broad category).
- c_D captures the smallest residual nuances (fine-grained item details).

E. The Visual Gap and Multimodal Encoders

The biggest shortcoming of current SID framework models like TIGER is their exclusive reliance on *text*. They only apply RQ-VAE to *text* embeddings. When dealing with highly visual categories (like fashion, home decor, or short-form video) - textual descriptions are often incredibly sparse or completely identical for items that actually look different.

To bridge this visual gap, we look at foundational visual-language models like CLIP (Contrastive Language-Image Pre-training) [Radford et al. 2021]. CLIP is trained with an InfoNCE contrastive loss, which mathematically forces the image's embedding to be highly cosine-similar to the embedding of its text description, and dissimilar to unrelated texts. CLIP provides a unified latent space where visual features can be mathematically quantified alongside text descriptions.

F. Recent Advances: 2025-2026 SOTA

State-of-the-art research has identified several bottlenecks in early GR models, leading to a wave of new advanced frameworks:

- **Multimodal Fusion (CEMG):** The CEMG framework addresses the superficial integration of collaborative signals by using them as queries to dynamically weigh visual and textual features during tokenization [Y. Lin et al. 2026].
- **Differentiable Indexing (DIGER):** To address the "objective mismatch" where SIDs are optimized for reconstruction rather than recommendation, DIGER implements Gumbel-Softmax reparameterization that allows recommendation gradients to flow back to the codebook [Fu et al. 2026].
- **Long ID Potential (ACERec):** ACERec decouples tokenization granularity from computational budget using an "Attentive Token Merger," allowing for expressive 32-token IDs while also maintaining efficient inference [Xia et al. 2026].

III. Problem Formulation and Gap Analysis

While foundational visual-language models like CLIP [Radford et al. 2021] bridge the visual-text gap, applying RQ-VAE to multimodal embeddings poses two critical limitations.

A. Limitation 1: Modality Sensitivity and Early Fusion Collapse

The intuitive approach to MGR is "Early Fusion": concatenating text and image embeddings before the RQ-VAE. Zhu et al. 2025 proved a flaw *'modality sensitivity'*. During

early fusion, one modality could mathematically dominate the representation space. Even recent dynamic weighing models like CEMG [Y. Lin et al. 2026] still operate within an early-fusion architecture, leaving them fundamentally vulnerable to cross-modal noise and indexing collisions for visually distinct items sharing similar text.

B. Limitation 2: The Semantic-Behavioral Gap

Xu et al. 2025 identified the *'Semantic-Behavioral Gap'*. An RQ-VAE trained solely on content assumes that items with similar aesthetics should receive similar SIDs. However, a designer black t-shirt and a budget black t-shirt, although visually identical, appeal and cater to entirely different user demographics. While advanced models like DIGER [Fu et al. 2026] attempt to bridge this by end-to-end differentiable indexing, their Gumbel-Softmax routing is extremely unstable and complex to train. A more modular and stable pipeline is required to align semantic generation with actual user preferences.

IV. Proposed Methodology: MASI

To resolve these bottlenecks, we propose the **Multimodal-Augmented Semantic ID (MASI)** framework. MASI consists of three distinct phases that are designed to maximize modality independence and behavioral alignment.

A. Phase 1: Behavior-Aware Contrastive Alignment

To solve the Semantic-Behavioral Gap (Limitation 2), our representations must respect collaborative user signals. However, if we mix the modalities while learning these signals, we risk modality interference, where the model ignores visual nuances in favor of text. To prevent this, we isolate the training. We freeze the base CLIP encoders so they do not suffer from catastrophic forgetting, and we attach independent, learnable projection heads to each modality. This allows each modality to learn user preferences independently. For an item i with image V_i and text T_i:

$$e_{vis,i} = \text{Proj}_v(\text{CLIP}_{vis}(V_i)), \quad e_{txt,i} = \text{Proj}_t(\text{CLIP}_{txt}(T_i)) \tag{2}$$

Using a user-item bipartite graph $\mathcal{G}$, we extract collaborative positive pairs (i, j^+). We then apply separate InfoNCE contrastive losses using **mixed negative sampling** $\mathcal{N}_{mix}$ (in-batch + hard graph negatives):

$$\mathcal{L}_{vis} = -\log \frac{\exp(\text{sim}(e_{vis,i}, e_{vis,j^+})/\tau)}{\sum_{k \in \mathcal{N}_{mix}} \exp(\text{sim}(e_{vis,i}, e_{vis,k})/\tau)} \tag{3}$$

The total alignment loss '$\mathcal{L}_{beh}$' is

$$\mathcal{L}_{beh} = \mathcal{L}_{vis} + \mathcal{L}_{txt}$$

Handling Cold-Start: When a brand-new item is added to the catalog, it has no user interaction history (no edges in G), meaning we cannot calculate the behavioral loss $\mathcal{L}_{beh}$ for it. To solve this, we simply pass the new item's raw text and image through our previously trained projection heads. Because these heads have already learned the general rules of how content maps to user preferences (using the older, "warm" items), the new items are automatically placed into the correct behavioral space without needing any prior clicks.

B. Phase 2: Independent Modality Quantization (Late Fusion)

To solve the problem of Modality Collapse (where text features often overpower visual features), we process the aligned text and image embeddings completely separately. By maintaining two entirely independent RQ-VAE pipelines, we ensure that the visual characteristics are matched to a dedicated visual codebook (C^{vis}), while the text characteristics are matched to a dedicated text codebook (C^{txt}). This guarantees that neither modality loses its unique information. For a given item i :

$$S_i^{vis} = \text{RQ-VAE}_{vis}(e_{vis,i}) = \langle c_1^v, c_2^v, \ldots, c_D^v \rangle \tag{4}$$

$$S_i^{txt} = \text{RQ-VAE}_{txt}(e_{txt,i}) = \langle c_1^t, c_2^t, \ldots, c_D^t \rangle \tag{5}$$

The final identifier is formed via Late Fusion with special modality tokens:

$$S_i^{final} = \texttt{[TXT]} \circ S_i^{txt} \circ \texttt{[VIS]} \circ S_i^{vis} \tag{6}$$

C. Phase 3: Recommender-Side Cross-Modal Alignment

As demonstrated by Zhu et al. 2025, utilizing separate codebooks does not inherently equip an autoregressive sequence model to comprehend cross-modal correlations. To bridge this, we introduce an auxiliary *pre-training* phase on the exact Transformer backbone used for downstream recommendations. Specifically, the model is trained to reconstruct visual tokens S_i^{vis} from textual tokens S_i^{txt} (and vice versa) using a cross-modal Masked Language Modeling (MLM) objective. After this discrete alignment phase, the Transformer is finally fine-tuned on chronological user interaction sequences. In essence, Phase 1 ensures that the dense, continuous vectors respect user behavior before they enter the codebooks, while Phase 3 ensures that the final, discrete SID tokens are cross-modally aligned for the autoregressive Transformer. We anticipate that these two stages operate synergistically (one organizes raw data and the other organizes the generative vocabulary) though we will conduct ablation studies to verify that they do not overlap in utility.

V. Theoretical Evaluation and Novelty

A. Comparative Novelty Analysis

MASI provides a streamlined alternative to recent SOTA paradigms (Table I).

MASI vs. CEMG: Although CEMG dynamically weights collaborative signals, it does so within a vulnerable early-fusion architecture. MASI proposes a collaboratively-aligned, yet architecturally separated, late-fusion approach, leading to modality independence during quantization.

MASI vs. DIGER: Compared to DIGER's elegant but unstable end-to-end differentiable indexing, MASI offers a pipeline-based alternative. Our staged approach (1. align continuous representations, 2. quantize independently, 3. align discrete sequences) is more modular, interpretable, and stable to train, while bridging the objective mismatch and the semantic-behavioral gaps.

Algorithm 1 MASI Generation & Alignment Pipeline

Input: User-Item Graph $\mathcal{G}$, Images V, Texts T
Phase 1: Behavior Alignment (Addressing Xu et al. 2025)
for batch $(i, j^+, \mathcal{N}^-)$ in $\mathcal{G}$ **do**
  $e_{vis} \leftarrow \text{Proj}_v(\text{CLIP}_{vis}(V))$, $e_{txt} \leftarrow \text{Proj}_t(\text{CLIP}_{txt}(T))$
  Compute $\mathcal{L}_{vis}$ and $\mathcal{L}_{txt}$ using graph positives and mixed negatives.
  Update Proj_v and Proj_t via backprop (CLIP remains frozen).
end for
Phase 2: Independent Quantization (Addressing Zhu et al. 2025)
$S^{vis} \leftarrow \text{RQ-VAE}_{vis}(e_{vis})$
$S^{txt} \leftarrow \text{RQ-VAE}_{txt}(e_{txt})$
$S^{final} \leftarrow \text{Concat}(\texttt{[TXT]}, S^{txt}, \texttt{[VIS]}, S^{vis})$
Phase 3: Sequence Pre-training & Fine-Tuning
Pre-train Transformer on MLM task: Predict S^{vis} from S^{txt}.
Fine-tune Transformer on user sequential interaction logs.
return Trained MASI Recommender System

TABLE I
ARCHITECTURAL NOVELTY COMPARISON

Framework	Fusion Type	Behavior Align	Rec-Alignment	Complexity
CEMG (2025)	Early	Dynamic Weight	None	High
DIGER (2026)	Text-Only	End-to-End	None	Very High
MGR-LF++	Late	None	Yes (Tokens)	Medium
MASI (Ours)	**Late**	**Proj. InfoNCE**	**Yes (Tokens)**	**Medium**

B. Mathematical Soundness of Dual Codebooks

Because we separate the quantization process (with the dual-codebook approach), the mathematical updates (gradients) for the visual loss ($\nabla \mathcal{L}_{RQ_vis}$) only ever affect the visual codebook (C^{vis}) and embedding (e_{vis}). By enforcing this strict architectural boundary, we mathematically prevent the text modality from overpowering the visual nuances during training. This directly resolves the modality collapse problem identified in early-fusion models.

VI. Experimental Design and Implementation Plan

A. Dataset Configuration

We will utilize the **Amazon Reviews 2023 Dataset** Lab 2023, filtering the "Clothing, Shoes, and Jewelry" subset (5-core strategy) to yield ∼100,000 active users and ∼50,000 items with valid high-resolution images.

B. Evaluation Protocol & Baselines

Employing a Leave-One-Out (LOO) protocol, we will partition the test set into *Warm-Start* and *Zero-Shot Cold-Start*. Metrics will include **HR@10**, **NDCG@10**, **Coverage@10** (percentage of catalog successfully recommended), and **Inference Latency (ms)**. Because tokenization is pre-computed, MASI's online inference latency matches MGR-LF++.

TABLE II
EXPECTED COMPARATIVE ANALYSIS TARGETS

Model	ID Type	Cold-Start HR	Cov@10	Lat.(ms)
SASRec	Atomic	0.000	12%	**15**
CEMG	Vis+Text	∼0.035	42%	85
MGR-LF++	Vis+Text	∼0.045	55%	65
DIGER	Text SID	∼0.050	52%	50
MASI	**Late+Beh.**	**>0.055**	**>65%**	**65**

C. Ablation Studies

We will conduct rigorous ablations to isolate contributions:

1) **Phase 1 vs. 3 Synergy:** MASI with vs. without Contrastive InfoNCE; with vs. without Recommender-side MLM to test relative redundancy.
2) **ID Length:** Varying RQ-VAE depth ($D \in \{2, 3, 4\}$).

D. Implementation Timeline (8 Weeks) & Feasibility

We will implement the dual-codebook quantization and contrastive graph sampling by leveraging modern AI-assisted coding agents (Google Gemini, GitHub Copilot) for rapid generation of code boilerplates (like PyTorch data-loader boilerplate and for parsing RecBole's internal abstractions).

- **Week 1-2:** Amazon 2023 ETL pipelines. Extract CLIP vectors.
- **Week 3-4:** Construct bipartite graphs; implement Phase 1 mixed negative sampling and train projection heads.
- **Week 5-6:** Train Phase 2 dual RQ-VAEs. Implement Phase 3 MLM pre-training within RecBole.
- **Week 7-8:** Execute generative fine-tuning, conduct LOO evaluations/ablations, and finalize the IEEE manuscript.

VII. Limitations and Future Work

MASI vs. ACERec: While MASI's late-fusion dual-codebook approach could prove to solve modality collapse, it doubles the generative sequence length per item ($2D$ tokens). We treat this as an accepted trade-off for capturing rich and independent multimodal representations. Future work will tackle the resulting linear increase in inference latency by incorporating ACERec's [Xia et al. 2026] Attentive Token Merging to compress the S^{final} representation prior to decoding, optimizing efficiency without sacrificing multimodal granularity.

VIII. Conclusion

Generative Recommendation's efficacy is bound by item representations. By going through 2025-2026 SOTA literature, we identified that basic multimodal extensions suffer from modality collapse and behavioral misalignment. The MASI framework tries to rectify these flaws through Independent Modality Quantization, Behavior-Aware Alignment,

and Cross-Modal Sequence Modeling, ensuring the generative engine benefits from true visual features and collaborative utilities to solve cold-start discovery.

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